



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Team 65: Solar Playground with Security System Bi-Weekly Update 5

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Kitchel, Victor Igbokwe
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TA: Zian Wang**

Project Summary

The SolarEye Security System is a solar-powered, autonomous surveillance solution for playgrounds and parks.

It enhances security with motion-activated cameras, real-time mobile alerts, LED deterrent lighting, and visitor charging stations.

This eco-friendly system ensures continuous, cost-effective protection for private parks and properties.



Integrated System Diagram (Hardware)



Figure 1: Full System Open Housing Diagram

Integrated System Diagram (Software)

Create an Account

Minimum 6 Characters, 1 Uppercase Character, 1 Special Character

Email Address

Password

Confirm Password

CREATE NEW ACCOUNT

BACK TO LOGIN

Figure 2: User Creates Account

Hello,

Follow this link to verify your email address.

https://solareye-security-system.firebaseio.com/_/auth/action?mode=verify

If you didn't ask to verify this address, you can ignore this email.

Thanks,

Your solareye-security-system team

Your email has been verified

You can now sign in with your new account

SolarEye

zayanprasla@tamu.edu

LOGIN

CREATE NEW ACCOUNT

Figure 4: User's email is verified and then able to login

Event History

LIVE STREAM

2025-03-26

Event Type: Motion Detected
7:23 PM

Charging Stats

Time	Charging Stats
09	90.0
15	80.0
21	75.0
03	60.0
09	55.0
15	70.0
21	85.0
03	95.0
09	100.0

LOW POWER MODE

SYSTEM CONTROLS

SETTINGS

Event History Page with live steam button and playable videos

Figure 5: Home page with updating Event thumbnail and low power mode and charging statistics

System Controls

Figure 6: System Controls for LED DETERRENCE, CAMERA, and SOS.

Camera Control

ON OFF

Light Control

ON OFF

Emergency Call

Are you sure you want to call 911?

NO YES



Project Timeline

New PCBs Ordered/ Received (completed 02/03)	Integration of MPU and Application subsystems (to completed 02/25)	Integration of Power Delivery system with MPU (to complete by 02/25)	Integration of Power Generation System with Power Delivery System (to complete by 03/26)	Final Integration (to complete by 04/10)	System Housing (to complete by 04/10)	Validation (to complete by 04/20)	Demo and Report (to complete by 04/25)
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Jackson Kitchel

Accomplishments since last update 9 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ Sensors have been connected via ADC to MCU ❖ Finished integration with Power Distribution Subsystem	❖ Finish setting up sensor algorithms in coordination with MCU subsystem ❖ Housing

Jackson Kitchel

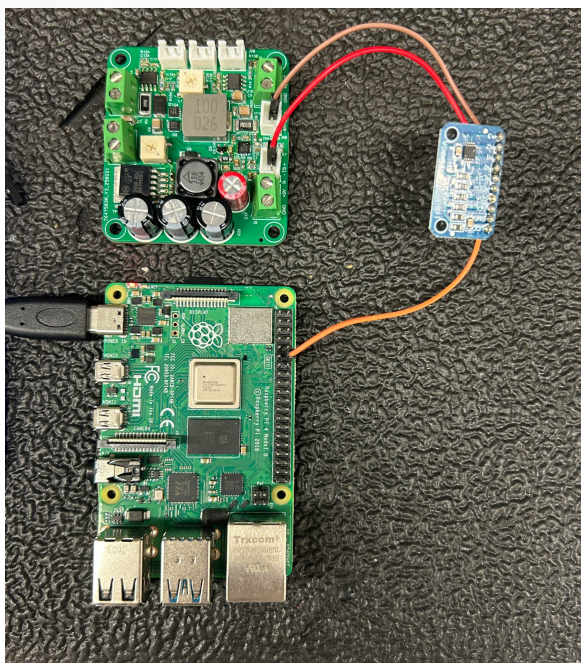


Figure 7: Sensor communication setup with ADC and MCU

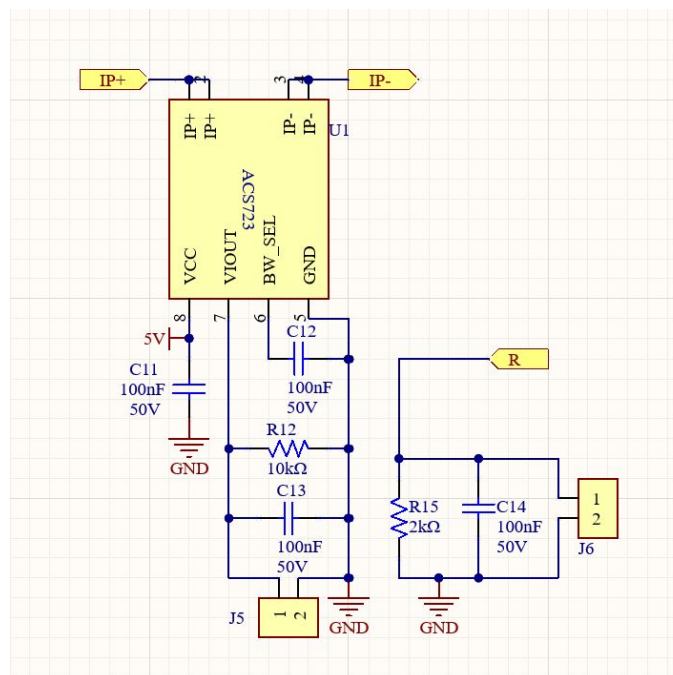


Figure 8: Sensor schematic built into PCB



Victor Igbokwe

Accomplishments since last update 13 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ Finished Integration with Power Generation Subsystem ❖ Created a Switchboard to improve safety when dealing with battery.	❖ Finish Up Housing Box ❖ House PCBs and other devices in box

Victor Igbokwe

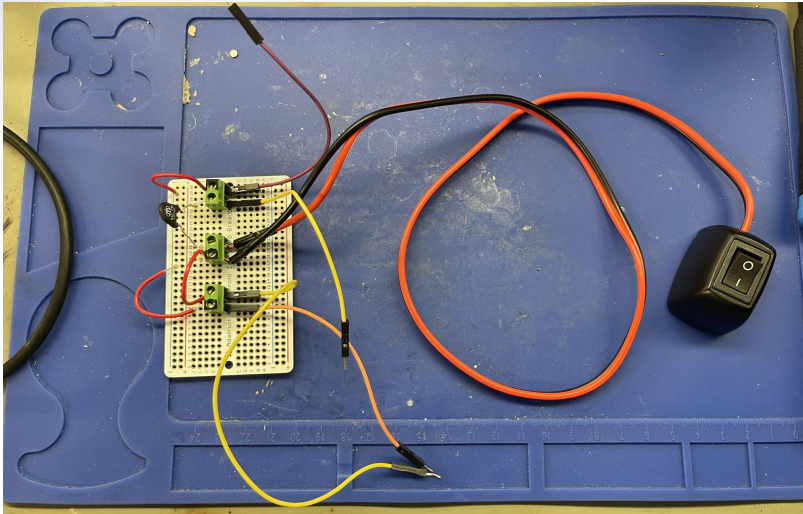


Figure 9: Switch board for protection



Figure 10: PCB Housing



Alex Allahar

Accomplishments since last update 12 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ Developed LED Deterrence Button for App ❖ Tested ADC I2C communication with Power Generation PCB	❖ Replace Camera ❖ Validate Charging Statistics ❖ Complete Housing

Alex Allahar

```
pi@raspberrypi1:~/sotareye $ cat adc_test.py
import time
import board
import busio
import adafruit_ads1x15.ads1115 as ADS
from adafruit_ads1x15.analog_in import AnalogIn

# Initialize the I2C bus
i2c = busio.I2C(board.SCL, board.SDA)

# Create the ADS1115 ADC instance
ads = ADS.ADS1115(i2c)

# Set the gain (adjust based on your input voltage range)
ads.gain = 1 # Available options: 1, 2/3, 1, 2, 4, 8, 16

# Create single-ended input channels for A0 and A1
chan_a0 = AnalogIn(ads, ADS.P0) # A0 corresponds to ADS.P0
chan_a1 = AnalogIn(ads, ADS.P1) # A1 corresponds to ADS.P1

while True:
    # Read the voltage values
    voltage_a0 = chan_a0.voltage
    voltage_a1 = chan_a1.voltage
    print(f'Voltage at A0: {voltage_a0:.2f} V')
    print(f'Voltage at A1: {voltage_a1:.2f} V')
    time.sleep(1)
```

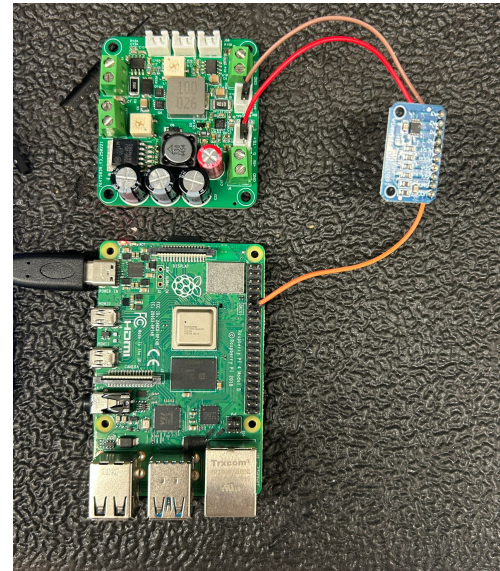


Figure 11: ADS 1115 Code / Setup



Zayan Prasla

Accomplishments since last update 11 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ Successfully validated video data transfer from MCU to app ❖ Updated event history thumbnail on home page to dynamically display the latest detection ❖ Implemented buttons for live streaming and low power mode functionalities	❖ Validate/test Low Power mode ❖ Validate/test Live Streaming ❖ Validate/test charging statistics

Zayan Prasla

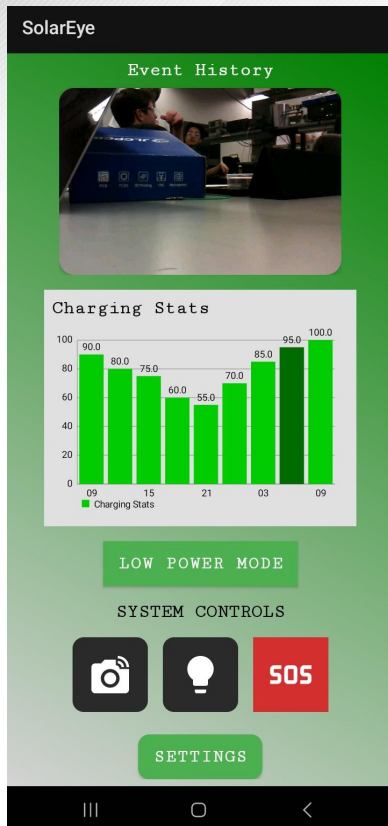


Figure 12: Main page shows latest event thumbnail & Low Power Mode Button

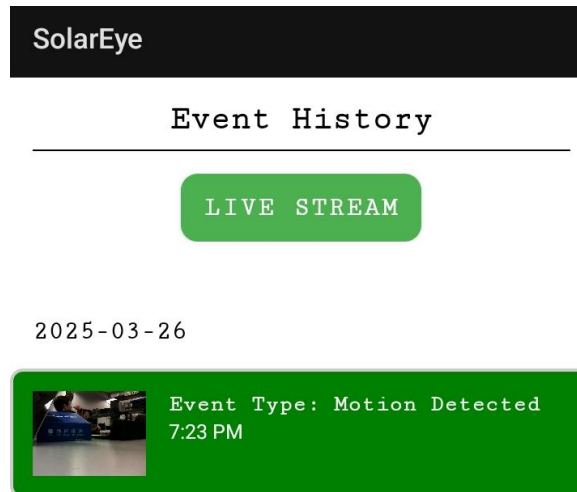


Figure 13: Event History with Live Stream button and updated event item thumbnail from Firebase Firestore



Figure 14: Video Playback screenshot

Execution Plan

[illegible]



SYSTEM Validation Plan Part 1

Test Name	Success Criteria	Methodology	Status	Assigned to
LED Relay Controller MPU Control	Be able to control the LED from on to off state, or automatically if motion is detected from the MPU.	Using CAM_validation.py to test if the USB Camera can send a 30 second video to Firebase, and determine time between video capture and firebase acknowledgement.	Completed	Alex Allahar
Modes of Operation App Control	Be able to control all three modes of operation via the app	Using firestore database and GCP functions to send a update to the MPU changing the mode of operation.	Completed	Alex Allahar
App Event History	Both MPU and App has up to a week of footage.	Verify that the MPU and Firebase after reboot has the up to 24 of prevent footage.	Completed	Alex Allahar
Video Streaming	The MPU live-stream can be access via App.	Use UDP to Stream the MPU via HTTP Flask Server.	Completed	Alex Allahar
Component App Control	Be able to control fans, LED, and camera from the App via Ffirebase.	Test modes of operation using comp_control.py .	Pending	Alex Allahar
MPU to App Automatic Delay	Delay is below 1 s. To make it almost automatic.	Using PIR_sensor_data_firestore.py perform 10 testcases and average the delay.	Completed	Alex Allahar
Charging Statistics to Firebase	Communicate in less that 1s the Charging Statistics from the Power Generation PCB.	Use GPIO pins to gather the current, and voltage data of the battery to determine the system uptime and communicate this information with Firestore.	Completed	Alex Allahar
Replace USB Camera with CPI Camera	Replace the system camera and ensure it is capable of running the modes of operation.	Setup Camera in integrated system and run the modes of operation, validating with firebase and app notifications.	Pending	Alex Allahar
Validate Buck integration with MPU	All modes of operation on MPU function while integrated wit buck converter PCB	Connect PCB and test all modes of operation	Completed	Victor Igbokwe
Buck LineTest	Output voltage remains constant with different input voltages	Change input voltage and record output voltages	Completed	Victor Igbokwe
Buck Load Test	Output voltage remains constant as the output current increases.	Use Electronic load device	Completed	Victor Igbokwe
Relay control module Integration	Current flow can be controlled by MPU	Connect GPIO pins on PCB to MPU	Completed	Victor Igbokwe



SYSTEM Validation Plan Part 2

Buck Converter integration with Solar Generation subsystem	Buck converter powered by solar generation system	Connect buck PCB to battery	Completed ▾	Victor Igbokwe
Network Connectivity	Able to receive and send signals to and from the app/system.	MPU data is able to be sent and seen on the app.	Completed ▾	Zayan Prasla
Password Policy	User has to put in a password that follows the password policy	Check if password can be created by not following the policy: 6 minimum	Completed ▾	Zayan Prasla
Store and Fetch Data	Application is able to store and fetch login data.	Test logging into the app with an already created account	Completed ▾	Zayan Prasla
Email Verification	After creating an account, user receives a verification email.	Check for a verification email being sent after creating an account	Completed ▾	Zayan Prasla
Main Page UI	Successfully display Event History, <i>Charging statistics</i> , and system controls	Check each function on main page and make sure there is no input lag	Partial Completion ▾	Zayan Prasla
Live Streaming	Live stream footage directly from the system camera to the app	Check if live stream is not too behind from actual live camera time	Pending ▾	Zayan Prasla
Low Power Mode	Turn on lower power mode with a switch on the app which communicates with the mpu	Check if mpu switches to low power mode once applied from app	Pending ▾	Zayan Prasla
Video/Picture Transfer	Successfully transfer pictures and videos with the correct date/time	Check the event history page and make sure picture and video align with date and time	Completed ▾	Zayan Prasla
Solar Generation Check	A charge rate of 0.45kW.	Test solar charge rate with a multimeter.	Completed ▾	Jackson Kitchel
Operation Time	8 hours of battery life	Monitor the battery after it finishes fulling charging.	Pending ▾	Jackson Kitchel
High Voltage Cut-Off	Charging circuit cuts off when voltage level reaches 15V	Monitor the charging voltage after battery is fully charged	Completed ▾	Jackson Kitchel
Low Voltage Cut-Off	Battery starts charging when low-voltage of 12.4V	Monitor the charging voltage after charge level is low	Completed ▾	Jackson Kitchel
PCB Functionality	PCB Soldered and functioning as required by previous checks	Validate every requirement after PCB is complete	Completed ▾	Jackson Kitchel
PCB Design	PCB Design is plausible.	Simulation Results using software	Completed ▾	Jackson Kitchel

Thank You



March 28, 2025
Night @ Zach